SkillMatch AI

Concept Note | Local-first Career Skill-Gap Analyzer

Project alignment

Primary: SDG 4 — Quality Education. Supporting: SDG 8 — Decent Work and Economic Growth. The project aims to make career skill requirements easier to interpret and turn into a learning plan.

WHY — Problem and motivation

Students preparing for internships and entry-level roles may find it difficult to compare their current skills with job descriptions. Requirements can be lengthy, unfamiliar, and hard to prioritize. A lightweight, transparent tool can organize this comparison and suggest next steps. These are problem hypotheses to validate, not claims about every student.

WHAT — Proposed solution

SkillMatch AI is a local-first Streamlit application. A user enters skills and experience, selects a target role, and may paste a job description. The app recognizes terms from a small local skill catalogue, compares profile text with role text using TF-IDF and cosine similarity, lists matched and missing skills relative to a curated role map, and provides a role-specific learning roadmap. Users can download a CSV summary and TXT roadmap.

HOW — Technical implementation

Python, Streamlit, pandas, and scikit-learn. Role and skill catalogues are bundled as JSON. Skill recognition uses case-insensitive phrase matching. TF-IDF with unigrams/bigrams and cosine similarity compares text. Skill coverage is the share of role-catalogue skills recognized in the profile. Roadmaps are curated locally. No large dataset, API key, account system, or runtime model download is required.

Supported roles

The initial catalogue covers Data Analyst, Machine Learning Intern, Data Scientist, Frontend Developer, and Backend Developer. This is a starting point, not a comprehensive job taxonomy. A pasted job description influences text similarity but does not produce a fully validated extraction of all requirements.

WHOM — Intended users

Primary: university students and early-career applicants exploring technology roles. Secondary: placement cells, career clubs, and mentors who want a structured discussion starter.

Outputs and interpretation

The app shows recognized skills, matched and missing catalogue skills, skill coverage, text similarity, and a learning roadmap. Text similarity measures language overlap; skill coverage measures catalogue matches. Neither is a hiring probability, proof of proficiency, or validated predictor of job performance. Users should verify suggestions against the original description.

Expected impact and evaluation

Evaluate functional correctness, catalogue recognition, user feedback, and roadmap usefulness. Suggested pilot metrics include task completion, user-rated usefulness, missed-skill rate found through manual review, and repeat use. No measured educational or employment impact is claimed before testing.

Feasibility and deployment

Designed for local Python 3.10+ (3.10–3.12 recommended). Deploy to Streamlit Community Cloud by pushing the repository to GitHub and selecting app.py. No secrets or external runtime network dependency. Vercel is not the

simplest host for Python Streamlit.

Risks and safeguards

The small catalogue may omit synonyms, emerging technologies, transferable skills, or role-specific requirements. Phrase matching can miss differently worded skills. Avoid entering sensitive personal data. The app must not rank candidates for hiring or imply a score guarantees employment.

Next steps

Run local tests, try multiple profiles and job descriptions, capture genuine screenshots, ask students or a placement mentor to review usefulness and errors, expand the catalogue using reviewed sources, and revise the pitch based on feedback. Add student name and institution before submission.

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